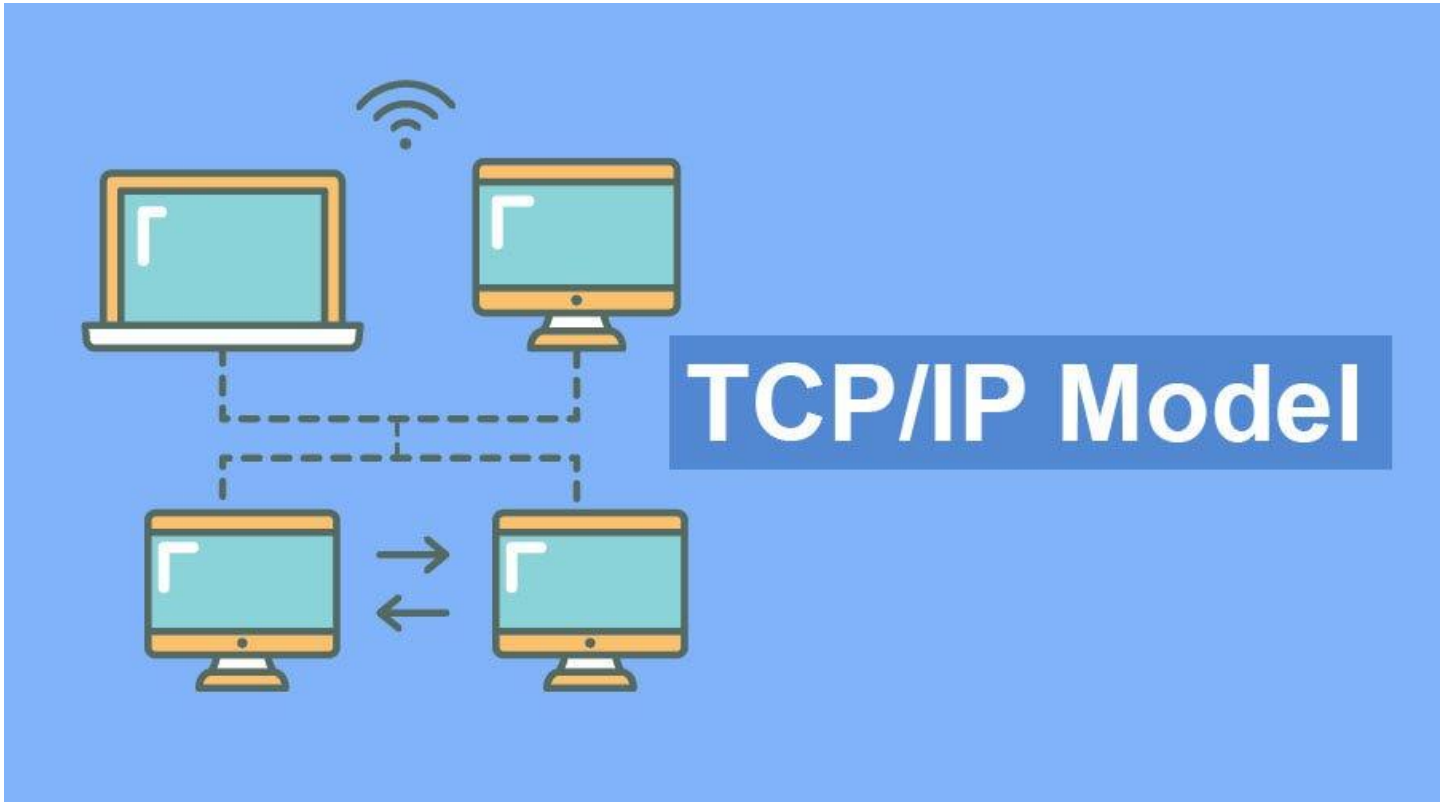
TCPIP Model



The TCP/IP Model

The Foundation of Communication on the Internet 🌐

What is TCP/IP?

- **TCP/IP** stands for **Transmission Control Protocol / Internet Protocol**
- It is a **set of rules (protocols)** that allows computers to communicate over networks.
- Used by **all devices** on the internet and local networks.
- Ensures **data is sent, delivered, and understood correctly.**

Why Do We Need TCP/IP?

Computers need standards to communicate. TCP/IP answers:

- How to **start** communication
- How to **send** data
- How to **address** where data goes
- How to **check** data has arrived safely
- How to **end** communication

➡️ Without TCP/IP, the internet would not work.

TCP vs IP

Protocol	Role	Think of it as…
IP	Finds the **address** to send data to.	The **address** on a mailing envelope.
TCP	Ensures the data is **delivered correctly** and **in order**.	A **delivery service** that checks every package.

The 4 Layers of the TCP/IP Model

1. **Application Layer**
2. **Transport Layer**
3. **Internet Layer**
4. **Network Access / Data Link Layer**

Data **moves through these layers** before being sent — and again when received.

Layer 1: Network Access / Data Link Layer

Handles **physical connection** and **hardware**.

- Network cables (Ethernet)
- Wi-Fi signals
- Network Interface Card (NIC)
- MAC Addressing

📍 *This is where data leaves the device.*

Layer 2: Internet Layer (Network Layer)

Handles **packet movement across networks**.

- **IP addresses**
- **Routing** (deciding best path across networks)
- Routers operate here

🌍 *This layer makes sure the packet goes to the correct destination network.*

Layer 3: Transport Layer

Ensures **data is delivered correctly**.

Tasks:

- Break data into **packets**
- Re-assemble packets at destination
- Ensure **all packets are received**
- Request re-send if needed

🔁 *Think of it as "reliable delivery tracking."*

Layer 4: Application Layer

This is the **layer users interact with**.

Examples:

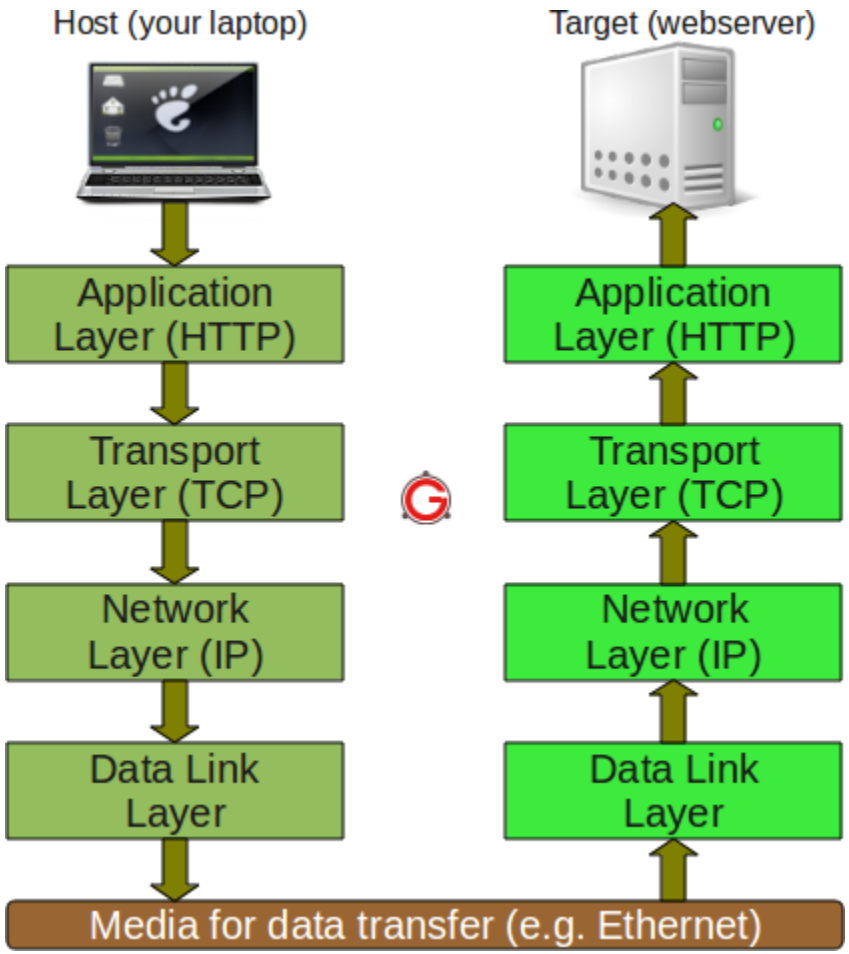
- Web Browsers 🌐 (HTTP / HTTPS)
- Email ✉️ (SMTP)
- File Transfer 📁 (FTP)
- Messaging Apps 💬

👩‍💻 *This is where data becomes meaningful to the user.*

TCP/IP Communication Flow

Your Computer → (Application → Transport → Internet → Data Link) → Network → Receiving Computer → (Data Link → Internet → Transport → Application)

Data **goes down the layers**, travels across the network, then **goes up the layers** at the destination.



TCP/IP vs OSI Model

TCP/IP Model	OSI Model
4 Layers	7 Layers
Practical, used in real networks	More theoretical
Simpler, modern	Used for teaching concepts